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- 1) The function of the protein can be anticipated by domain homology searches with various confidence level.

2) atgaacctgcaggcgcagccgaaagcgcagaaacaaacgcaaacgctgcctgtttggcggc
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gaagaagtggaagtggatattaaaaccagccagaaatttccgcgcgtgccgctgccgcgc

3) a) Steps of Cloning

Isolation of the DNA of interest

Ligation

Transformation or transfection

Screening/ selection

need detailed plan

b) An expression vector has different features that vector have like origin of replication, site for insertion of gene and selectable maker. The cloning site is performed in *Escherichia coli*.

c) T7 Terminator: GCTAGTTATTGCTAGCGG

T7 Universal primer: TAATACGACTCACTATAGGG

4) The host cell would be *Escherichia coli*

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